Building Valid, Credible, and Appropriately Detailed Simulation Models

Systems Modeling and Simulation

4DV650

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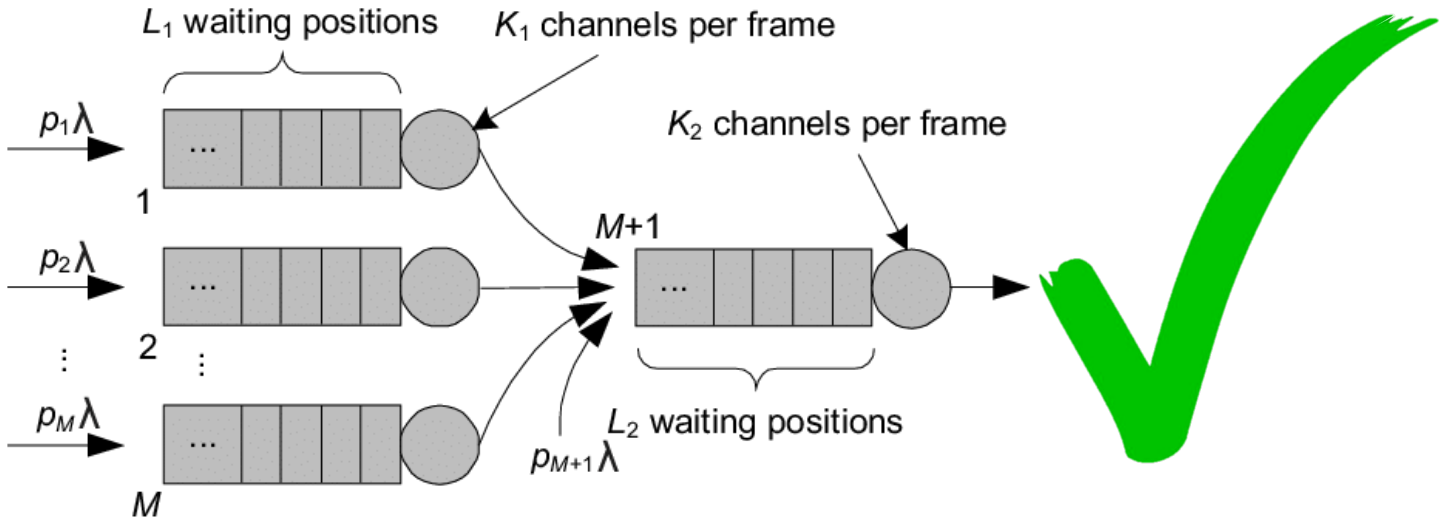
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Problem statement

- Is the simulation model a **valid** and **accurate representation** of the actual system being studied?



1. A simulation model **can only approximate** the actual system
2. The output should include what the decision maker really needs for **evaluating system designs**

Verification, Validation, Credibility, Accreditation

- **Verification** is concerned with determining whether the "assumptions document" has been correctly translated into a computer "program"
  - Did we build the **model right**?
  - i.e., **debugging** the simulation computer program
- **Validation** is the process of determining whether a simulation model is an accurate representation of the system, *for the particular objectives of the study*
  - i.e., Did we build the **right model?**
- **Credibility** is supported by key project personnel
  - A model and its results have **credibility** if they are accepted as "correct"
  - A credible model is not necessarily valid, and vice versa
- **Accreditation** is the official certification that a simulation model is acceptable for a specific purpose

Validation and Output Analysis

Validation should be contrasted with **output analysis**

- Concerned with estimating a simulation model's **true measures of performance**
  - e.g., simulation run length, number of independent model runs, etc.

Difference between validation and output analysis

- Suppose we want to estimate the **mean** μ_S of a given system
- Suppose we construct a simulation model with (theoretical) **mean** μ_M

$$\begin{aligned}\text{Error in } \hat{\mu}_M &= |\hat{\mu}_M - \mu_S| && \hat{\mu}_M \text{ estimate of } \mu_M \\ &= |\hat{\mu}_M - \mu_M + \mu_M - \mu_S| \\ &\leq |\hat{\mu}_M - \mu_M| + |\mu_M - \mu_S| && \text{(by the triangle inequality)}\end{aligned}$$

Output analysis aims at making this value small

Validation aims at making this value small

Guidelines for determining the level of model detail

- Models are not universally valid, but are **designed for specific purposes**
  - It is important to understand the **real needs**: a great model for the wrong problem will never be used
- An entity in the simulation does not have to be the **same** as the entity of the corresponding system
  - It is not always necessary to model each component of the system in **complete detail**
- Use **Experts** and **Sensitivity Analysis** to help determine the level of model detail
  - Experts are familiar with the system and **know** what components need to be modeled
  - Sensitivity analysis allows for determining what system **factors** have the greatest impact on the desired measures of performance
    - e.g., parameters or distributions

Guidelines for determining the level of model detail (contd)

- Do not have more detail in the model than is **necessary** to address the issue of interest
- Start with a **moderately detailed** model, and embellish it later, if needed
- The level of model detail should be consistent with the type of **data available**
- Use a “coarse” simulation model or an analytic model to identify what **factors** have a significant impact on system performance
  - A “detailed” simulation model is then built, emphasizing these factors

Verification of simulation computer programs - Techniques

1. Write and **debug** the computer program in modules or subprograms
   - Write and debug the simulation model's main program and a few of the key subprograms
2. More than one person to **review** the computer program
3. Run the simulation under a **variety of settings**
   - Run the simulation under a variety of settings of the input parameters
   - Simple measures may be computed exactly (from theory) and used for comparison
4. Trace a program
   - Display the state of the simulated system just after each event occurs
     - i.e., the event list, the state variables, statistical counters, etc.,
   - Compare it with hand calculations to see if the program is operating as intended
5. The model should be run under **simplifying assumptions**
   - True characteristics are known or can easily be compute

Techniques for increasing model validity

1. Collect **high-quality** information and data on the system
   - Conversation with Subject-Matter Experts
   - Observation of the System
   - Existing Theory
2. Interact with the manager on a **regular basis**
   - The manager's knowledge contributes to the actual validity of the model
   - The model is more credible since the manager understands and accepts the assumptions
3. Maintain an **assumptions document**
   - Includes Model concepts, assumptions, algorithms, and data summaries

4. Validate Components by Using Quantitative Techniques

- Quantitative techniques should be used whenever possible to **test the validity** of various components of the overall model

- An important technique for determining which model **factors** have a significant impact on the desired measures of performance is **sensitivity analysis**
  - e.g., a **parameter's** value, a **distribution**, moving **entity,**

- One-factor-at-a-time (OFAT)
  - One of the most used approaches to see what **effect** this produces on the output
    1. Moving **one input variable**, keeping others at their **baseline** (nominal) values, then,
    2. Returning the variable to its nominal value, then **repeating for each of the other inputs**
  - Sensitivity may be measured by **monitoring changes** in the output
    - Any change observed in the output is **unambiguously** due to the single variable changed

5. Validate the Output from the Overall Simulation Model

- Establish that model's output resemble the output expected from the actual system

- If a similar system exists, compare output data with data from the system itself
- In alternative
  - Use a **Turing Test**
  - Ask for **Expert Opinion**
  - Compare with **another model**

Materials (MyMoodle)

- Reading
  - S2 [Lecture] Building Valid Credible and Appropriately Detailed Simulation Models.pdf

- Exercises (with solutions):
  - S2 [Lecture] Building Valid Credible and Appropriately Detailed Simulation Models - Exercises.pdf
  - S2 [Lecture] Building Valid Credible and Appropriately Detailed Simulation Models - Solutions.pdf